DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #03B: MARTIAL ARTS & TECHNIQUES (PART 2 — COMPOSABLE ENGINE, MULTI-TIER ELEMENTAL FUSION, ERA-5 SPECIALIST ARTS, PASSIVE CONSTELLATIONS & SYNERGIES)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Companion Document**: Read alongside **`03A_JS2_Deep_Audit_Martial_Arts_Part1_Combat_Moves_Architecture_Unlocking_AI_and_Spatial_Execution`** (which covers active move generation, `SK.list` caching, `chooseSkill` scoring, weapon mismatches, and spatial range bugs).
- **Baseline Filter Note (Crucial for AI Coders)**: All 16 issues from `JS2_v2.4.0_Deep_QA_Techniques_Report_TH` (`TECH-P0-01`..`TECH-P2-16`) were already patched in `v2.4.0 Release (1)` (`JS2_V2_4_0_FIX_TRACKER_TH.md`). **Every finding in this document (`[TECH-ELEM-09]` through `[TECH-SYN-14]`) is a newly uncovered or still-live defect verified directly against the current `JS2-v2.6.0` source and test suites (`v235`, `v340`, `v540`–`v580`).**

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0. Hard Architectural & Design Invariants to Preserve

1. **Immutable Human Core**: Never modify `GAME_SOURCE/js/human_core.js` (`SHA-256: a0d393284fb4b136848e5b80e12f3edd2c4f9437c419c793c2a7e75d881c0542`, `HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **40-Script Load Order**: Preserve exact script load sequence in `GAME_SOURCE/index.html` and `GAME_SOURCE/test/load.js`.
3. **100% Emergent NPC Autonomy (Zero Player Puppeteering)**: Players are sandbox observers/editors; they **never** micro-command NPCs in combat or sect operations. Every stance, aura, seal mode, constellation focus, sect barrier tune, formation stance, and emergency invocation must be driven autonomously by NPCs and appointed Sect Officers according to `HC` personality, intellect, combat context, and resource reserves.

4. **Zero-RNG Read Purity & Tick Budget**: Pure evaluators (`STATS.get`, `SK.list`, `Render.draw`) must never mutate world state or advance `W.rng`. Keep 80-NPC `SIM.tick` $\le 3.5\text{ ms/tick}$.

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1. [TECH-ELEM-09] The DARK_YIN vs YIN_SHADOW Schema Schism & Elemental Gauge Starvation

1.1 Exact Code Defect: 6 Yin-Based Multi-Tier Reactions Are 100% Unreachable in Real Gameplay

- **Locations**:
  - `js/combat.js` (`COMBAT.COMPOSABLE_ENGINE.DUAL_REACTIONS`, `TRI_REACTIONS`, `TETRA_REACTIONS`)
  - `js/human_core.js` (`HC.AFF_TH`), `js/cultivation.js` (`ELEM.ELEMENTS`), `js/economy.js` (`ALL_ELEMENTS`), `data/data_tech.js`, `data/data_martial.js`
  - `test/v235_comprehensive_mechanics_and_elemental_fusion_suite.js:163–195`
- **Mechanism of Failure**:
  - Across the entire character, item, root, subpath, and technique catalog, the Yin/Shadow element is canonically keyed as **`'YIN_SHADOW'`** (ธาตุหยิน-เงา). Zero techniques, zero weapons, zero spirit roots, and zero subpaths emit `'DARK_YIN'`.
  - However, when the 34-reaction Multi-Tier Elemental Fusion Matrix (`16 Dual + 12 Tri + 6 Tetra`) was authored in `js/combat.js`, the Yin element was hardcoded as **`'DARK_YIN'`** in **6 major reactions**:
    1. `2E-11 TAIJI_ANNIHILATION` (`LIGHT_YANG + DARK_YIN`)
    2. `3E-07 COSMIC_ECLIPSE` (`ASTRAL + LIGHT_YANG + DARK_YIN`)
    3. `3E-08 VOID_THUNDER_MIRAGE` (`LIGHTNING + WIND + DARK_YIN`)
    4. `3E-11 STYGIAN_FREEZE` (`WATER_ICE + BLOOD_DEMON + DARK_YIN`)
    5. `4E-03 SAMSARA_TWILIGHT` (`LIGHT_YANG + DARK_YIN + BLOOD_DEMON + ASTRAL`)
    6. `4E-05 NETHERWORLD_ROOT` (`WOOD_LIFE + WATER_ICE + DARK_YIN + BLOOD_DEMON`)
  - **Why Existing Tests Missed It (False-Positive Test Trap)**: In `test/v235_comprehensive_mechanics_and_elemental_fusion_suite.js:168–193`, the test loops over `spec.elements` from `CE.DUAL_REACTIONS` / `TRI_REACTIONS` / `TETRA_REACTIONS` and feeds `spec.elements[0]` (`'DARK_YIN'`) directly back into

`CE.applyElementalAura` and `CE.triggerElementalReaction`! Because the test tested the reaction table against itself rather than casting an actual `YIN_SHADOW` technique (`t.element === 'YIN_SHADOW'`), all 401 assertions passed while every Yin fusion remained completely dead in live gameplay!

- **Cascading Cross-System Impact**:
  1. **Combat**: `YIN_SHADOW_HARMONY`, `SHADOW_ASSASSIN`, `NETHER_GHOST_KING`, `NINPO_SHADOW`, and `NETHER_GRAVE` cultivators can never trigger Taiji Annihilation, Stygian Freeze (`healMultiplier = 0`), Cosmic Eclipse, or Samsara Twilight.
  2. **Elemental Rune Smithing (`ECO.socketElementalRunes`) & Alchemy (`ECO.brewMultiphaseElementalPill`)**: Socketing a `YIN_SHADOW` rune alongside `LIGHT_YANG` fails to embed `2E-11`, and brewing Yin-Yang multiphase pills fails to resolve the Yin-Yang zero-toxicity fusion.
  3. **Secret Realm Elemental Gates (`EXPLORE.solveElementalPuzzleGate`)**: Any puzzle gate requiring one of the 6 Yin reactions cannot be solved by disciples holding `YIN_SHADOW` roots, manuals, or weapons unless normalized.

1.2 Elemental Gauge Starvation & AI Blindness to Active Elemental Auras

- **Locations**: `js/combat.js:480–520` (`C.strike`), `js/combat.js:686–689` (`chooseSkill`), `js/combat.js:1164` (`decayElementalAuras`)
- **Mechanism of Failure**:
  1. **Gauge Starvation for 3E & 4E Reactions**: `decayElementalAuras` drains `0.2U` per tick on every active combat tick, while normal elemental techniques and elemental weapons (`atkElem !== 'NEUTRAL'`) do **not** deposit Primer Gauge Units (`0.8U–1.0U`) unless the move explicitly carries a `v234/v235` aura tag. Consequently, accumulating 3 simultaneous elemental auras on a boss to detonate a 4-Element (`4E`) reaction almost never happens naturally even in a diverse 5-person party.
  2. **AI Scoring Blindness (`chooseSkill` line 687)**:

```JavaScript
if (p.arch === 'elementalFusion') {
  const hasReactable = d.sx && d.sx.some(x => ['burn', 'chill', ...].includes(x.k)
&& x.until > W.t);
  sc += hasReactable ? 2.15 : 0.55;
}
```

`chooseSkill` checks legacy `d.sx` statuses instead of `CE.getActiveElementalAuras(d, W.t)`! Even `GRAND_TACTICIAN` casters do not check whether casting a move of element E will complete a `2E`, `3E`, or `4E` fusion with the auras currently sitting on `d._elemAuras`.

1.3 Remediation Blueprint ([TECH-ELEM-09])

- Create a canonical `CE.normalizeElementKey(el)` helper mapping `'DARK_YIN' -> 'YIN_SHADOW'` (and `'ICE' -> 'WATER_ICE'`, `'METAL' -> 'EARTH_METAL'`, `'WOOD' -> 'WOOD_LIFE'`, `'SHADOW' -> 'YIN_SHADOW'`, `'LIGHT' -> 'LIGHT_YANG'`, `'BLOOD' -> 'BLOOD_DEMON'`) across `DUAL_REACTIONS`, `TRI_REACTIONS`, `TETRA_REACTIONS`, `applyElementalAura`, `triggerElementalReaction`, `socketElementalRunes`, `brewMultiphaseElementalPill`, and `solveElementalPuzzleGate`.
- Deposit a modest primer aura (`0.85U` on elemental skill hit, `0.35U` on elemental weapon basic hit, respecting per-attacker-element ICD) so multi-disciple parties naturally build toward `2E/3E/4E` reactions.
- In `chooseSkill`, evaluate `CE.previewElementalReaction(d, p.element, W.t)`: award `+1.15` for completing a `2E` reaction, `+2.10` for `3E`, and `+3.20` for `4E` (weighted by `intellect.comprehension` / `GRAND_TACTICIAN`).

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2. [TECH-COMP-10] Residual Composable Action Bugs, Reflection Cap Bypass & O(N) Aura Scans

2.1 Broken Composable Action Implementations (MARIONETTE_SOUL_SWAP & KHMER_YANTRA_WARD)

- **Location**: `js/combat.js` (`COMBAT.COMPOSABLE_ENGINE.executeAction`)
- **Defects**:
    1. **`MARIONETTE_SOUL_SWAP` (`SOUL_PUPPETRY` signature action)**: Tethers target `t` with `linkedToId: t.id` (pointing to **itself**) when no explicit secondary target is passed, instead of selecting a second living enemy (`C.enemies(W, f, actor.id).find(e => e.id !== t.id)`). Self-tethering makes the soul-link damage split a complete no-op.
    2. **`KHMER_YANTRA_WARD` (`OCCULT_MAGE` signature action)**: Calls `ST.applyNamedBuff(actor, 'ironSkin', ...)`—but `'ironSkin'` is **not registered** in `ST.BUFF_CATALOG` (`js/stats.js:116`) nor in `SK.STATUS`. `ST.applyNamedBuff` returns `null`, granting **0 DEF, 0 MDEF, and 0 shield**! Map it to `'aegisWard'` or register `'ironSkin'` in `ST.BUFF_CATALOG` (`{ p: { def: 0.24, mdef: 0.18 }, shieldPct: 0.16 }`).

2.2 Multiple Uncapped Damage Reflections Bypassing `COMBAT.reflectOf` (`V240-08` Regression)

- **Location**: `js/combat.js:244–420` (`C.applyDamage`) vs `js/combat.js:518` (`C.strike -> C.reflectOf`)
- **Mechanism of Failure**:
  - Although `V240-08` consolidated keyword/buff reflection into `C.reflectOf` (capped at $\le 35%$ of incoming damage and $\le 22%$ of attacker MaxHP), **six separate subsystems inside `C.applyDamage` still fire raw, uncapped `C.applyDamage` reflection hits BEFORE `C.reflectOf` even runs**:
    1. `BLACK_TORTOISE` Spirit Pet (`line 256`): reflects `22%` of intercepted damage.
    2. `VAJRA_IMMORTAL_BODY_BELL` Mantra (`line 283`): reflects `16%` of incoming damage.
    3. `SWORD_QI_MERIDIAN` Mantra (`line 290`): counter-slashes `12%` of incoming damage.
    4. `PRIMORDIAL_YANG_FURNACE` Mantra (`line 294`): burns back `8%` of incoming damage.
    5. `TAIJI_SOFT_REDIRECTION` / `ELEMENTAL_THORN_ARMOR` Defensive Arts (`line 339–346`): redirects `18%–32%` of **`rawBefore` (pre-mitigation raw damage!)**.
    6. `PARRY_COUNTER` Buff (`line 407`): reflects `48%` of defender ATK.
  - And then `C.strike` (`line 518`) adds **up to another `35%`** via `C.reflectOf`!
- **Chain-Reaction Impact**: A defensive cultivator stacking `VAJRA_IMMORTAL_BODY_BELL` + `TAIJI_SOFT_REDIRECTION` + `BLACK_TORTOISE` + `karmicAegis` reflects **90%–110%+ of pre-mitigation boss damage on a single hit** with **zero `22%` attacker MaxHP ceiling**, causing high-ATK bosses or burst assassins to instantly blow themselves up on contact.
- **Fix**: Accumulate all reactive/reflective damage sources per hit into a single per-hit reflection budget (`hitCtx.reflectedSoFar`) enforced by the unified ceiling `min(dmg * 0.40, C.hpMax(a) * 0.22)` (and base `TAIJI_SOFT_REDIRECTION` on post-armor `dmg`, not unmitigated `rawBefore`).

2.3 Post-Mortem Poison/DoT Kill Attribution Erasure (`C.statusTick`) & Past-Life Echo Desync

- **Locations**: `js/combat.js:956–960` (`C.statusTick`) & `js/combat.js:907` (`useSkill` Past-Life Echo)
- **Defects**:
  1. In `C.statusTick`: `C.applyDamage(W, f, src && !src.dead ? src : null, u, ...)` strips `src` to `null` if the poisoner/bleeder `src` died 1 tick earlier in a mutual trade. Passing `null` erases `src` from `C.slay`, preventing

`BLACK_THREAD` vengeance on `u`'s sectmates, skipping `recordWeaponKill` / `kills++` on `src`'s historical chronicle record, and misreporting the death cause as generic `"สิ้นชีพจากพิษหรือบาดแผลในการต่อสู้"` instead of naming the poisoner! Keep `killer = src` for attribution in `C.slay`, guarding only living-only rewards (`if (!killer.dead) { CULT.gain(...); ... }`).

2. In `useSkill` (`line 907`): `PAST_LIFE_MIRROR` echo strike targets original `d` instead of `actualT` (after bodyguard/LoS interception) and ignores redirected `selectTargets` (such as `LOWEST_HP_ENEMY`).

2.4 `O(N_npcs)` Full-Sect Loops Inside `C.applyDamage` (`combat.js:312–375`)

- **Location**: `js/combat.js:312` (`CELESTIAL_AEGIS_DOME`) and `js/combat.js:351, 363` (`CULT.SUPPORT_ARTS`)
- **Defect**: On **every single damage instance** (including multi-hit skills, pet assists, DoT ticks, and AoE hits across 6 targets), `C.applyDamage` iterates over the entire `W.npcs` array (`for (let i = 0; i < W.npcs.length; i++)`, up to 80–100 NPCs, most of whom are farming or sleeping across the map!) instead of iterating over `C.alliesOf(W, f, d.id)` (typically 1–5 combatants in the actual fight `f`).
- **Fix**: When `f` is an active fight with `f.A` / `f.B` (or `f.units`), iterate strictly over `C.alliesOf(W, f, actor.id)` (`O(party_size)` instead of `O(80)`), cutting raid CPU overhead in `applyDamage` by $>90%$.

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3. `[TECH-ERA5-11]` The Five Era-5 Specialist Art Systems (`v5.4–v5.8`) Violate Autonomous Simulation Rules

3.1 Systematic Diagnosis: Rich Engines Wired Exclusively to Player UI Buttons

Across Patches `v5.4.0` through `v5.8.0`, five major martial subsystems were added on top of `JD.TECH`:

1. **`CULT.QINGGONG` (`v5.4.0`)**: 7 locomotion/flight modes (`MORTAL_WALK` .. `SWORD_FLIGHT`, `CELESTIAL_LEVITATION`).
2. **`CULT.MANTRA_SYSTEM` (`v5.5.0`)**: 122 mantras across 7 Dharma archetypes + 3 stances (`SERENE_PURIFICATION`, `HARMONIOUS_CIRCULATION`, `OVERDRIVE_COMBUSTION`) + `SECT.FORMATION_WARFARE` (3 sect-wide formation stances).

3. **`CULT.DEFENSIVE_ARTS` (`v5.6.0`)**: 58 defensive techniques across 6 archetypes + 3 guard postures (`PASSIVE_AURA_GUARD`, `COUNTER_PARRY_STANCE`, `FORTRESS_BASTION_LOCK`).
4. **`CULT.SUPPORT_ARTS` (`v5.7.0`)**: 97 support/auxiliary techniques across 6 archetypes + 3 postures + `triggerSectEmergencySupport`.
5. **`CULT.FORBIDDEN_ARTS` & `CULT.PASSIVE_RESONANCE` (`v5.8.0`)**: 37 forbidden arts (3 seal modes: `SEALED_RESTRAINT`, `HALF_UNSEALED_FLOW`, `BLOOD_TABOO_UNLEASHED`), **263 passives/innates grouped into 6 Passive Constellations**, and `SECT.FORBIDDEN_VAULT` (Vault Policies & Bloodline Altar Rituals).

Critical Autonomy & Logic Failures Verified in `JS2-v2.6.0`:

1. **100% Dormant Passive Constellations (`n.passiveResonanceState`)**: All 263 passive and innate techniques carry `passiveConstellation`, and `C.applyDamage` (`combat.js:380–387`) + `CULT.rate` check `n.passiveResonanceState`. **However, zero NPCs ever initialize or update `n.passiveResonanceState` autonomously!** Unless the player opens every single NPC's sheet (`UI.rNpc`) and manually clicks a constellation focus button, **100% of NPCs in the world have `n.passiveResonanceState === undefined` and receive 0% Constellation Tier 1/2/3 bonuses forever.**
2. **Sect-Level War & Altar Controls Hidden in `UI.tab === 'visuals'` Without Officer AI**:
   - `SECT.FORMATION_WARFARE.setFormationStance` (`SWORD_NET_SLAUGHTER` / `IMMORTAL_AEGIS_DOME`),
   - `SECT.ELEMENTAL_BARRIER.setBarrierElement` & `rechargeBarrier`,
   - `COMBAT.invokeAncestralPhantom`,
   - `CULT.SUPPORT_ARTS.triggerSectEmergencySupport`, and
   - `SECT.FORBIDDEN_VAULT.setVaultPolicy` & `performAltarRitual` are all rendered as player-clickable buttons inside the Sect UI (`UI.rSect`), violating the core design rule that **players cannot command sect operations directly—NPC officers must govern the sect!** Meanwhile, the `arraymaster`, `healer`, `discipline` elder, and `master` never invoke these functions during raids, plagues, or heart-demon outbreaks.
3. **Zero Autonomous Stance/Posture/Seal Adaptation on Individual NPCs**:
   - An NPC with `toxic >= 35` or `heartDemon >= 40` never switches their `mantraState.stance` to `SERENE_PURIFICATION` during `A.cultivate`, nor switches back to `HARMONIOUS_CIRCULATION` / `OVERDRIVE_COMBUSTION` once purified.
   - An NPC learning a `DEFENSIVE_TECHNIQUE` or `SUPPORT_TECHNIQUE` never initializes `n.defensiveArtState` or `n.supportState`! Notice `W._hasActiveDefSupportAura` stays falsy unless manually toggled via UI, meaning the entire group healing, toxicity cleansing

(`stepFacilityAndDailySynergy`), and Aegis ally interception pipeline never activates organically!

- A righteous `discipline` elder never enforces `FORBIDDEN_VAULT` policy against disciples who unseal `BLOOD_TABOO_UNLEASHED` inside an orthodox sect (`CHARTER_SCHOLASTIC_DAO` / high `strictness`), whereas a desperate or demonic disciple (`DEMON` / high `ambition` / `hp < 30%` in lethal `serious: true` combat) never autonomously unseals their learned Forbidden Art to survive!

3.2 Remediation Blueprint (`[TECH-ERA5-11]`)

1. **Auto-Align Passive Constellation in `COG.reconcileLoadout` & `M.reconcilePassives`**:
   - Count learned + active passives/innates per `passiveConstellation`; automatically call `CULT.PASSIVE_RESONANCE.setFocusConstellation(W, n, bestConstellation)` whenever an NPC's passive loadout or stage changes (tie-breaking by subpath/personality).
2. **Auto-Initialize & Adapt `mantraState`, `defensiveArtState`, `supportState`, `qinggongState`, and `forbiddenArtState`**:
   - When an NPC awakens or learns a Mantra / Defensive / Support / Qinggong / Forbidden technique, initialize the corresponding state object automatically.
   - Evaluate posture transitions deterministically at activity start (`A.cultivate.start`, `A.supportBless.start`, `COMBAT.enter`, and `COMBAT.fightStep` at `<30% HP` in `f.serious === true` fights) using `HC` traits (`caution`, `aggression`, `empathy`, `integrity`, `ambition`) and vows (`VOW_ASCETIC_PRECEPT` strictly forbids unsealing unless `willpower` breaks under near-death trauma).
3. **Delegate Sect-Wide Tactical Buttons to Authorized Officers (Convert UI to Read-Only Status or Sandbox Override)**:
   - **`arraymaster` / `master`**: On raid start (`EXPED.spawnRaid` / `COMBAT.tick`), switch `FORMATION_WARFARE` stance (`IMMORTAL_AEGIS_DOME` if outnumbered or `caution >= 55`; `SWORD_NET_SLAUGHTER` if stronger and `aggression >= 60`), counter-attune `ELEMENTAL_BARRIER` against the attacker's primary element, and recharge the barrier when `W.sect.stones > cashReserve + 150`.
   - **`master` / `deputy`**: In a `raid` fight when Main Hall breach ticks > 15 or ≥ 2 defenders fall below $25%$ HP and `incensePoints >= 100`, autonomously call `COMBAT.invokeAncestralPhantom(W, f)`.
   - **`healer`**: During daily office duty, if ≥ 3 disciples suffer `heartDemon >= 45` or severe injuries, trigger `triggerSectEmergencySupport` (with proper Qi/resource cost and cooldown).

- **`master` / `discipline`**: Set `FORBIDDEN_VAULT.policy` organically from Sect Charter and Master alignment (`STRICT_SEAL` for orthodox/high-integrity masters; `EMERGENCY_ONLY` for pragmatic; `UNRESTRICTED_BLOOD_CULT` for `DEMON` / `VOID_HERETIC_PACT` masters).

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4. [TECH-PASS-12] Passive Loadout Authority Split, Magic Conditional Blindness & Broken-Gear Cache Leak

4.1 Competing Sorters Between `M.reconcilePassives` and `COG.reconcileLoadout`

- **Locations**: `js/martial.js:244–295` (`M.reconcilePassives`) vs `js/cognition.js:358–395` (`COG.reconcileLoadout`)
- **Mechanism of Failure**:
  - `M.reconcilePassives(n)` fills open slots purely by `(tb.tier - ta.tier) || (tb.power - ta.power)` **without checking `M.conditionMatches(n, t)`**! And once `n.passiveLoadout` is filled (`3/3` slots), `M.reconcilePassives` preserves existing cross-family passives indefinitely.
  - Consequently, if a Sword Immortal learns three Tier-4 Staff/Bow passives (or loots them from crystals), `M.reconcilePassives` slots all three unusable passives (`conditionMatches === false`, granting **0 stats**!), blocking compatible Tier-3 Sword passives until `COG.reconcileLoadout` runs.
  - Worse: `COG.reconcileLoadout`'s cache key `lSig` (`cognition.js:360`) checks `n.equip.WEAPON.id` (template ID) instead of `uid + ':' + !!M.gearUsable(n, eqW)`. If a procedural weapon breaks mid-day (`dur <= 0`, `M.gearUsable === false`) or is swapped for another procedural weapon of the same template `ITEM_PROC_SWORD`, `lSig` **does not change**, leaving dead weapon-locked passives slotted!
- **Fix**: Teach `M.reconcilePassives` itself to prioritize `M.conditionMatches(n, t) ? 1 : 0` ahead of raw `tier`, and include `(eqW ? (eqW.uid || eqW.id) + ':' + M.gearUsable(n, eqW) : '')` (and same for `ARMOR`) in both `_passRec`, `_passStatsCache`, and `_reconcileLoadoutSig`.

4.2 `M.combatTargetBonus` Ignores `cnd.p.matk` on All Qi/Spellcaster Conditional Passives!

- **Location**: `js/martial.js:383–406` (`M.combatTargetBonus`)
- **Exact Code**:

```javascript
if (cnd.p && cnd.p.atk) dmgMult += cnd.p.atk;
if (cnd.requiresDebuffOnTarget && dmgMult === 1) dmgMult += 0.08;
```

- **Mechanism of Failure**:
  - Spellcaster, Talisman, Zither, Venom, and Soul passives define target-conditional bonuses using `cnd.p.matk` (e.g., `{ targetHpBelowPct: 40, p: { matk: 0.24 } }` or `{ requiresDebuffOnTarget: true, p: { matk: 0.22 } }`).
  - Because `combatTargetBonus` checks **only `cnd.p.atk`**, every pure-`matk` target-conditional passive loses its entire `0.24` scaling (dropping to `0` on HP-threshold passives, or `0.08` fallback on debuff-target passives)!
  - Furthermore, in `C.strike` (`combat.js:483`), `tb.dmgMult` multiplies **both** physical (`type === 'p'`) and magical (`type === 'm'`) attacks indiscriminately, so a physical `cnd.p.atk` passive unintentionally boosts pure `matk` spells too!
- **Fix**: Pass attack `type` (`'p'` or `'m'`) into `M.combatTargetBonus(a, d, type)` (or return `{ physDmgMult, magDmgMult }`), reading `cnd.p.atk` when `type === 'p'` and `cnd.p.matk` when `type === 'm'`, plus reading `cnd.f` (`pen`, `crit`, `critDmg`) alongside `cnd.pp`.

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5. [TECH-SYN-13] Duet Skills (PROP-06), Equipment-Granted Skills & Resonance Set Edge Cases

5.1 Duet Skill Execution in Friendly Spars & Across Disabled/Distant Partners

- **Location**: `js/combat.js:619–628` (`C.fightStep` -> `C.tryExecuteDuetSkill`)
- **Defects**:
  1. **Fires Before Normal Skills in Friendly Spars/Tourneys**: `fightStep` attempts `tryExecuteDuetSkill` *before* `C.attack` whenever `(!n._duetCdUntil || W.t >= n._duetCdUntil)`, without checking `f.serious` or whether `al` is incapacitated (`stun`, `paralyze`, `silence`, `cryoStasis`, `channeling`) or out of realistic co-casting range (`C.unitDist(n, al) <= 4.5`) and has sufficient Qi! Two stunned or silenced companions standing 30 tiles apart in a raid can still fire a combined Duet Resonance nuke for 0 Qi.
  2. **Zero Spatial Cohesion Between Bonded Duet Partners**: If `n` is a `SHORT_SWIFT` melee fighter and their `companionId` is a `RANGED_KITE`

caster/archer, they never coordinate spacing when a Duet skill is off cooldown, causing distance-checked duets to fizzle by 0.5 tiles.

- **Fix**: Require both `n` and `al` to be free of hard CC/silence, within `4.0` tiles of each other, within range of `t`, and paying `8% qiMax` (or `15 pathGauge` for `SHADOW_SAMURAI`); restrict full-power Duets to `f.serious || f.kind === 'raid'` (or scale `×0.45` in nonlethal team drills).

5.2 Equipment-Granted Skills (`grantedSkills`) & Resonance Set Rider Scaling

- **Locations**: `js/martial.js` (`M.grantedEquipmentTechs`), `js/skills.js` (`SK.list`), `js/combat.js:522–538`
- **Defects**:
  1. **Granted Active Skills Bypass Path/Elemental Identity Without Penalty**: When a pure `BODY` or `BUDDHA_DHARMA` monk equips a looted `EQ_WPN_V234_*` weapon that grants a Demonic/Forbidden skill in `item.grantedSkills`, `chooseSkill` scores and casts it without checking `VOW_ASCETIC_PRECEPT` or applying a Heart-Demon/karma friction penalty to incompatible righteous personalities (`integrity >= 70`).
  2. **3-Piece Resonance Set Rider DoTs Ignore Tier & Stats (`combat.js:529`)**:

```JavaScript
if (rd.status === 'burn' || rd.status === 'bleed' || rd.status === 'poison') {
  ST.addStatus(tgt, rd.status, rd.dur || 10, { per: 0.012, src: a.id });
}
```

Whether a cultivator wears a **Tier-3** Resonance Set or an endgame **Tier-7 Pagoda Set (`avgTier = 7`, `tierScale = 1.48`)**, the 3-piece rider DoT is hardcoded to `per: 0.012`, never scaling with `resSet.tierScale`! Scale `per` by `0.010 * (resSet.tierScale || 1)`.

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6. Phased Implementation & Verification Checklist for GPT / Gemini

Phase 1 — Schema & Stat-Pipeline Accuracy (Zero Balance Spike Risk)

- [ ] Normalize `'DARK_YIN'` $\leftrightarrow$ `'YIN_SHADOW'` across all 34 reactions in `COMBAT.COMPOSABLE_ENGINE`, `ECO.socketElementalRunes`, `ECO.brewMultiphaseElementalPill`, and

`EXPLORE.solveElementalPuzzleGate`. Write an end-to-end test casting an actual `YIN_SHADOW` technique + `LIGHT_YANG` technique to trigger `2E-11 TAIJI_ANNIHILATION` and `3E-11 STYGIAN_FREEZE`.

- [ ] Fix `MARIONETTE_SOUL_SWAP` secondary enemy targeting and `KHMER_YANTRA_WARD` buff registration (`ST.BUFF_CATALOG`).
- [ ] Fix `M.combatTargetBonus` (`js/martial.js:399`) to read `cnd.p.matk` for `'m'` attacks and `cnd.p.atk` for `'p'` attacks, plus `cnd.f`.
- [ ] Unify passive sorting in `M.reconcilePassives` (`conditionMatches` first) and fix `COG.reconcileLoadout` cache signature to include `uid` + `M.gearUsable`.

Phase 2 — Combat Engine Safety & CPU Optimization

- [ ] Route all 6 reactive reflection sources in `C.applyDamage` (`BLACK_TORTOISE`, `VAJRA_IMMORTAL_BODY_BELL`, `SWORD_QI_MERIDIAN`, `PRIMORDIAL_YANG_FURNACE`, `DEFENSIVE_ARTS`, `PARRY_COUNTER`) through a shared per-hit reflection accumulator capped at $\le 40%$ incoming damage and $\le 22%$ attacker MaxHP.
- [ ] Replace `for (i = 0; i < W.npcs.length)` scans inside `C.applyDamage` with `C.alliesOf(W, f, unit.id)`.
- [ ] Preserve `killer = src` on post-mortem DoT kills in `C.statusTick` (`combat.js:959`) so `C.slay` records accurate killer chronicles and `BLACK_THREAD` karma while guarding living-only stat gains with `if (!killer.dead)`.
- [ ] Add CC/silence, range (`<= 4.0` tiles), and Qi/stamina gates to `tryExecuteDuetSkill`.

Phase 3 — Full Emergent Autonomy for Era-5 Systems (`v5.4–v5.8`)

- [ ] Auto-compute and assign `CULT.PASSIVE_RESONANCE.setFocusConstellation` whenever an NPC's passive loadout updates.
- [ ] Auto-initialize `mantraState`, `defensiveArtState`, `supportState`, `qinggongState`, and `forbiddenArtState` upon learning matching techniques, and add deterministic personality-driven posture/seal transitions in `ai.js` and `combat.js`.
- [ ] Wire `arraymaster`, `healer`, `discipline`, and `master` officers in `sect.js` / `combat.js` to autonomously manage `FORMATION_WARFARE`, `ELEMENTAL_BARRIER`, `invokeAncestralPhantom`, `triggerSectEmergencySupport`, and `FORBIDDEN_VAULT` policies.
- [ ] Run `node test/core_lock.js`, `node tools/validate_data_schema.js`, `node tools/run_parallel_tests.js`, and `node tools/build_single.js`.